

1

Задача для самостоятельной работы

$$B_1 = \begin{bmatrix} 2 & -1 \\ 3 & 3 \end{bmatrix}$$

Найти d_0^+ , d_1^+ , d_2^+ , d_0^- , d_1^- , d_2^-

D_{00} , D_{11} , D_{22} , D_{01} , D_{02} , D_{12} , D_{21}

ещё надо

проверить $D_{12} = D_{21}$

первая точка: $(0, 0)$

вторая точка: $(2, 3)$

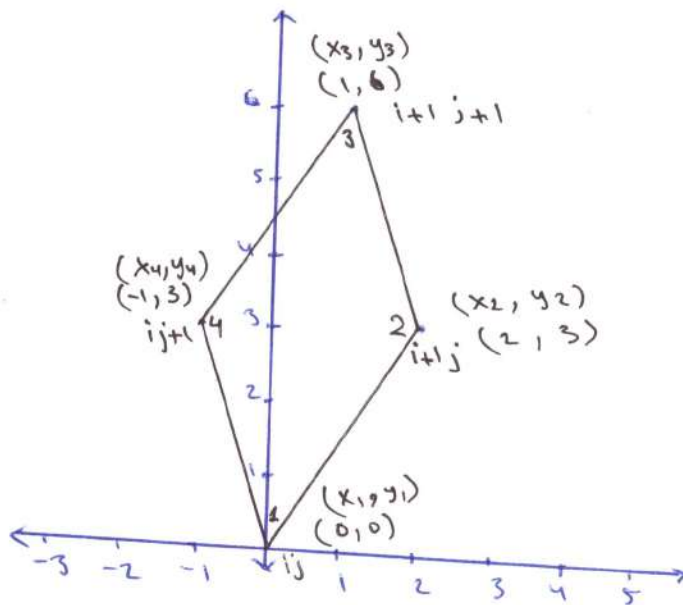
третья точка: $(-1, 3)$

четвертая точка: $(2 + (-1), 3 + 3) = (1, 6)$

$$d_0 \sim I$$

$$d_1^+ \sim \frac{\partial}{\partial x}$$

$$d_2^+ \sim \frac{\partial}{\partial y}$$



точка 1 = точка 5

$$(d_0^+ f)_{ij} = \frac{1}{4} [f_{ij} + f_{i+1,j} + f_{i+1,j+1} + f_{i,j+1}]$$

$$\begin{aligned} (d_1^+ f)_{ij} &= \frac{\sum_{i=1}^4 f_i (y_{i+1} - y_{i-1})}{\sum_{i=1}^4 x_i (y_{i+1} - y_{i-1})} = \frac{f_1 (y_2 - y_4) + f_2 (y_3 - y_1) + f_3 (y_4 - y_2) + f_4 (y_1 - y_3)}{x_1 (y_2 - y_4) + x_2 (y_3 - y_1) + x_3 (y_4 - y_2) + x_4 (y_1 - y_3)} \\ &= \frac{f_1 (3 - 3) + f_2 (6 - 0) + f_3 (3 - 3) + f_4 (0 - 6)}{0 (3 - 3) + 2 (6 - 0) + 1 (3 - 3) + (-1) (0 - 6)} \\ &= \frac{1}{18} [6f_2 - 6f_4] \\ &= \frac{1}{3} [f_2 - f_4] \end{aligned}$$

$$(d_1^+ f)_{ij} = \frac{1}{3} [f_{i+1,j} - f_{i,j+1}]$$

2

ТОЗКА 0 = ТОЗКА 4
ТОЗКА 5 = ТОЗКА 1

$$\begin{aligned}
 (d_2^+ f)_{ij} &= \frac{\sum_{i=1}^4 f_i (x_{i+1} - x_{i-1})}{\sum_{i=1}^4 y_i (x_{i+1} - x_{i-1})} \\
 &= \frac{f_1 (x_2 - x_4) + f_2 (x_3 - x_1) + f_3 (x_4 - x_2) + f_4 (x_1 - x_3)}{y_1 (x_2 - x_4) + y_2 (x_3 - x_1) + y_3 (x_4 - x_2) + y_4 (x_1 - x_3)} \\
 &= \frac{f_1 (2 - (-1)) + f_2 (1 - 0) + f_3 (-1 - 2) + f_4 (0 - 1)}{(0)(2 - (-1)) + (3)(1 - 0) + 6(-1 - 2) + 3(0 - 1)} \\
 &= -\frac{1}{18} [-3f_1 + f_2 - 3f_3 - f_4] \\
 &= \frac{1}{18} [-3f_1 - f_2 + 3f_3 + f_4]
 \end{aligned}$$

$$(d_2^+ f)_{ij} = \frac{1}{18} [-3f_{ij} - f_{i+1j} + 3f_{i+1j+1} + f_{ij+1}]$$

$$(d_0^+ f)_{ij} = \frac{1}{4} [f_{ij} + f_{i+1j} + f_{i+1j+1} + f_{ij+1}]$$

$$(d_0^- f)_{ij} = \frac{1}{4} [f_{ij} + f_{i-1j} + f_{i-1j-1} + f_{ij-1}]$$

$$(d_1^+ f)_{ij} = \frac{1}{3} [f_{i+1j} - f_{ij+1}]$$

$$(d_1^- f)_{ij} = \frac{1}{3} [f_{i-1j} - f_{ij-1}]$$

$$(d_2^+ f)_{ij} = \frac{1}{18} [-3f_{ij} - f_{i+1j} + 3f_{i+1j+1} + f_{ij+1}]$$

$$(d_2^- f)_{ij} = \frac{1}{18} [-3f_{ij} - f_{i-1j} + 3f_{i-1j-1} + f_{ij-1}]$$

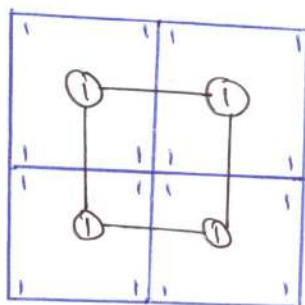
3

D_{00} :

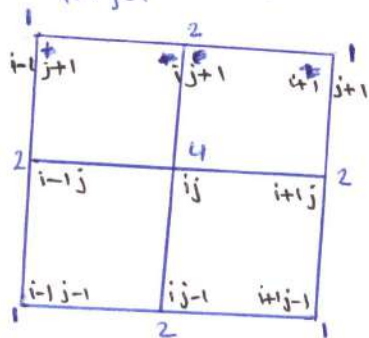
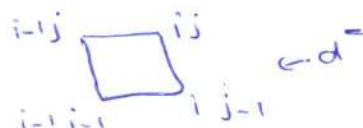
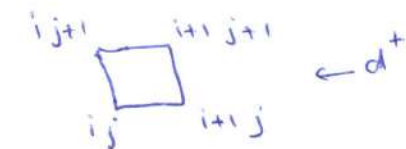
$$(d_0^+ f)_{ij} = \frac{1}{4} [f_{ij} + f_{i+1,j} + f_{i,j+1} + f_{i+1,j+1}]$$

$$(d_0^- f)_{ij} = \frac{1}{4} [f_{ij} + f_{i-1,j} + f_{i,j-1} + f_{i-1,j-1}]$$

$$(D_{00} f)_{ij} = d_0^+ d_0^- = \frac{1}{4} \cdot \frac{1}{4}$$



$$= \frac{1}{16}$$



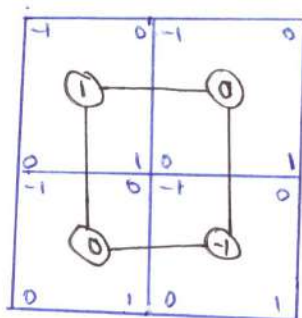
$$(D_{00} f)_{ij} = \frac{1}{16} [f_{i-1,j+1} + 2f_{i,j+1} + f_{i+1,j+1} + 2f_{i-1,j} + 4f_{i,j} + 2f_{i+1,j} + f_{i-1,j-1} + 2f_{i,j-1} + f_{i+1,j-1}]$$

D_{11} :

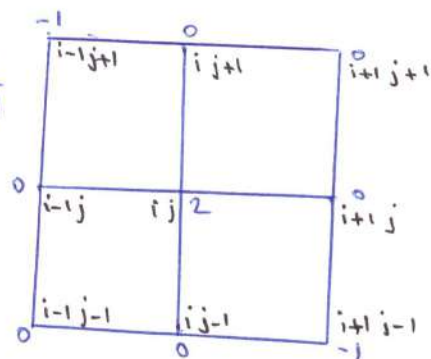
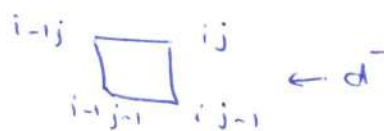
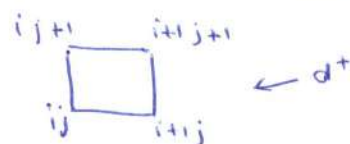
$$(d_1^+ f)_{ij} = \frac{1}{3} [f_{i+1,j} - f_{i,j+1}]$$

$$(d_1^- f)_{ij} = \frac{1}{3} [f_{i-1,j} - f_{i,j-1}]$$

$$(D_{11} f)_{ij} = d_1^+ d_1^- = \frac{1}{3} \cdot \frac{1}{3}$$



$$= \frac{1}{9}$$



$$(D_{11} f)_{ij} = \frac{1}{9} [-f_{i-1,j+1} + 2f_{i,j} - f_{i+1,j-1}]$$

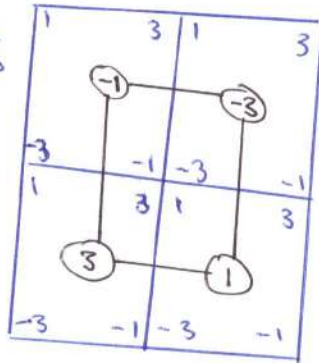
4

D_{22} :

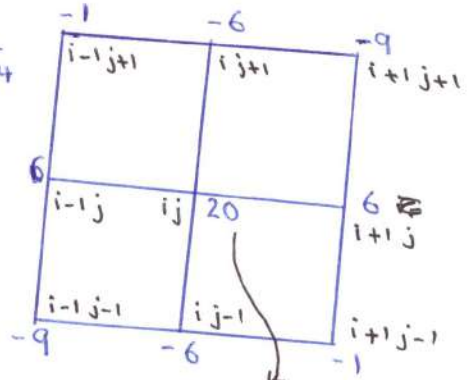
$$(d_2^+ f)_{ij} = \frac{1}{18} [-3 f_{ij} - f_{i+1j} + 3 f_{i+1j+1} + f_{i+1j+1}]$$

$$(d_2^- f)_{ij} = \frac{1}{18} [-3 f_{ij} - f_{i-1j} + 3 f_{i-1j-1} + f_{i-1j-1}]$$

$$(D_{22} f)_{ij} = d_2^+ d_2^- = \frac{1}{18} \cdot \frac{1}{18}$$



$$= \frac{1}{324}$$



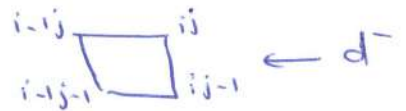
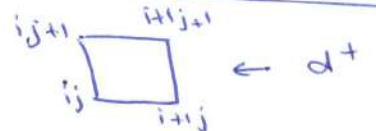
$$20 = (-3)(-3) + (-1)(-1) + (3)(3) + (1)(1)$$

$$(D_{22} f)_{ij} = \frac{1}{324} [-f_{i-1j+1} - 6 f_{ij+1} - 9 f_{i+1j+1} + 6 f_{i-1j} + 20 f_{ij} + 6 f_{i+1j} - 9 f_{i-1j-1} - 6 f_{ij-1} - f_{i+1j-1}]$$

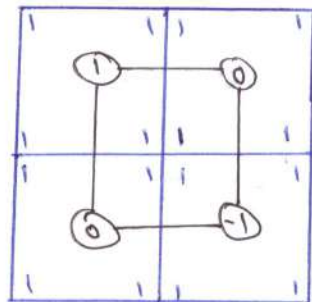
D_{01} :

$$(d_0^+ f)_{ij} = \frac{1}{4} [f_{ij} + f_{i+1j} + f_{i+1j+1} + f_{i+1j+1}]$$

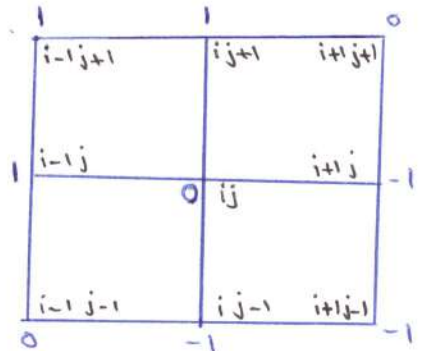
$$(d_0^- f)_{ij} = \frac{1}{3} [f_{i-1j} - f_{i+1j-1}]$$



$$(D_{01} f)_{ij} = d_0^+ d_0^- = \frac{1}{4} \cdot \frac{1}{3}$$



$$= \frac{1}{12}$$



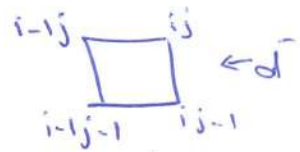
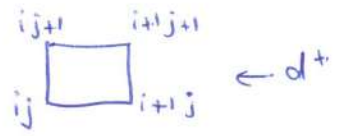
$$(D_{01} f)_{ij} = \frac{1}{12} [f_{i-1j+1} + f_{ij+1} + f_{i+1j+1} - f_{i-1j} - f_{i+1j} - f_{i+1j-1}]$$

5

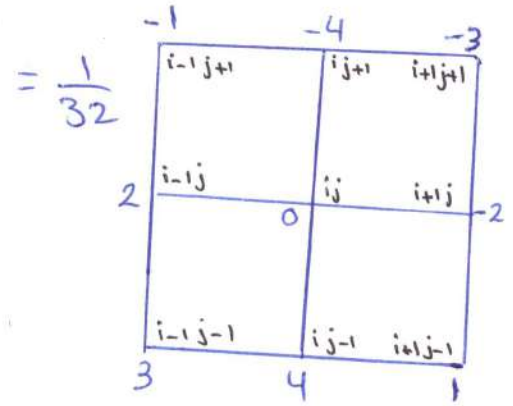
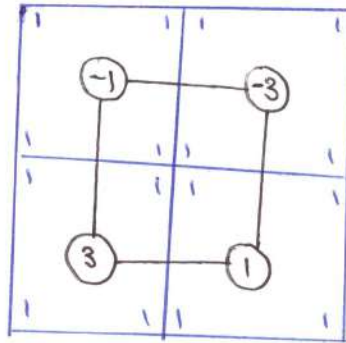
D_{02} :

$$(d_0^+ f)_{ij} = \frac{1}{4} [f_{ij} + f_{i+1j} + f_{i+1j+1} + f_{ij+1}]$$

$$(d_0^- f)_{ij} = \frac{1}{8} [-3f_{ij} - f_{i-1j} + 3f_{i-1j-1} + f_{ij-1}]$$



$$(D_{02} f)_{ij} = d_0^+ d_0^- = \frac{1}{4} \cdot \frac{1}{8}$$

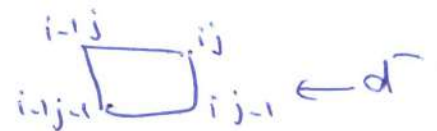
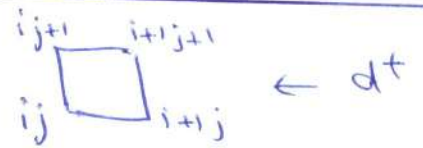


$$(D_{02} f)_{ij} = \frac{1}{32} [-f_{i-1j+1} - 4f_{ij+1} - 3f_{i+1j+1} + 2f_{i-1j} - 2f_{i+1j} + 3f_{i-1j-1} + 4f_{ij-1} + f_{i+1j-1}]$$

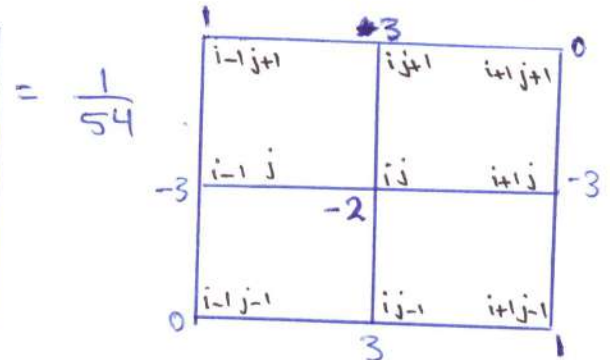
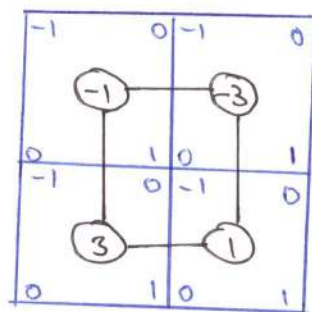
D_{12} :

$$(d_1^+ f)_{ij} = \frac{1}{3} [f_{i+1j} - f_{ij+1}]$$

$$(d_1^- f)_{ij} = \frac{1}{18} [-3f_{ij} - f_{i-1j} + 3f_{i-1j-1} + f_{ij-1}]$$



$$(D_{12} f)_{ij} = \frac{1}{3} \cdot \frac{1}{18}$$

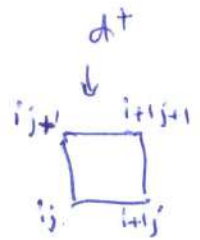


$$(D_{12} f)_{ij} = \frac{1}{54} [f_{i-1j+1} + 3f_{ij+1} - 3f_{i-1j} - 2f_{ij} - 3f_{i+1j} + 3f_{ij-1} + f_{i+1j-1}]$$

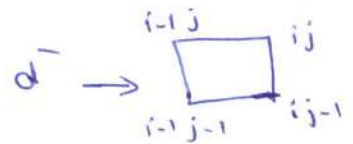
6

$D_{21} :$

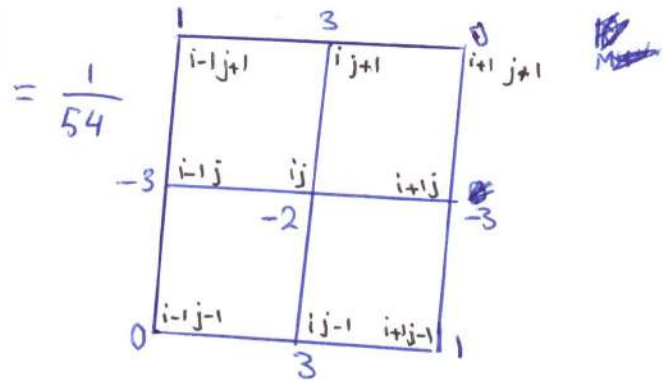
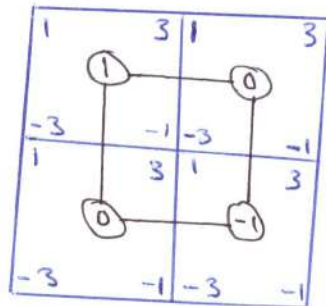
$$(d_2^+ f)_{ij} = \frac{1}{18} [-3f_{ij} - f_{i+1,j} + 3f_{i+1,j+1} + f_{i,j+1}]$$



$$(d_1^- f)_{ij} = \frac{1}{3} [f_{i-1,j} - f_{i,j-1}]$$



$$(D_{21} f)_{ij} = \frac{1}{18} \cdot \frac{1}{3}$$



$$(D_{21} f)_{ij} = \frac{1}{54} [f_{i-1,j+1} + 3f_{ij+1} - 3f_{i-1,j} - 2f_{ij} - 3f_{i+1,j} + 3f_{i,j-1} + f_{i+1,j-1}]$$

$\Downarrow$

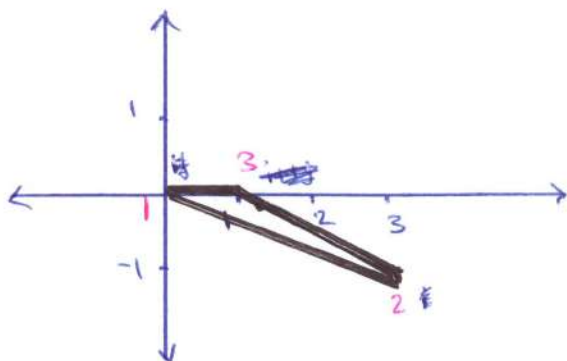
$$(D_{12} f)_{ij} = \frac{1}{54} [f_{i-1,j+1} + 3f_{ij+1} - 3f_{i-1,j} - 2f_{ij} - 3f_{i+1,j} + 3f_{i,j-1} + f_{i+1,j-1}]$$

$$\Rightarrow (D_{12} f)_{ij} = (D_{21} f)_{ij}$$

7

$$B_2 = \begin{bmatrix} 3 & 1 \\ -1 & 0 \end{bmatrix}$$

Найти $D_{00}, D_{11}, D_{22}, D_{01}, D_{10}, D_{02}, D_{20}, D_{12}, D_{21}$



первый узел: $(0,0) = (x_1, y_1)$

второй узел: $(3,-1) = (x_2, y_2)$

третий узел: $(1,0) = (x_3, y_3)$

$$(d_0^+ f)_{ij} = \frac{1}{3} [f_{ij} + f_{i+1,j} + f_{ij+1}]$$

$$\begin{aligned} (d_1^+ f)_{ij} &= \frac{\sum_{i=1}^3 f_i (y_{i+1} - y_{i-1})}{\sum_{i=1}^3 x_i (y_{i+1} - y_{i-1})} = \frac{f_1 (y_2 - y_3) + f_2 (y_3 - y_1) + f_3 (y_1 - y_2)}{x_1 (y_2 - y_3) + x_2 (y_3 - y_1) + x_3 (y_1 - y_2)} \\ &= \frac{f_1 (-1 - 0) + f_2 (0 - 0) + f_3 (0 - (-1))}{0 (-1 - 0) + 3 (0 - 0) + 1 (0 - (-1))} \\ &= [-f_1 + f_3] \end{aligned}$$

$$(d_1^+ f)_{ij} = [-f_{ij} + f_{ij+1}]$$

$$\begin{aligned} (d_2^+ f)_{ij} &= \frac{\sum_{i=1}^3 f_i (x_{i+1} - x_{i-1})}{\sum_{i=1}^3 y_i (x_{i+1} - x_{i-1})} = \frac{f_1 (x_2 - x_3) + f_2 (x_3 - x_1) + f_3 (x_1 - x_2)}{y_1 (x_2 - x_3) + y_2 (x_3 - x_1) + y_3 (x_1 - x_2)} \\ &= \frac{f_1 (3 - 1) + f_2 (1 - 0) + f_3 (0 - 3)}{(0)(3 - 1) + (-1)(1 - 0) + (0)(0 - 3)} \\ &= [2f_1 + f_2 - 3f_3] \end{aligned}$$

$$(d_2^+ f)_{ij} = 2f_{ij} + f_{i+1,j} - 3f_{ij+1}$$

8

$$(d_0^+ f)_{ij} = \frac{1}{3} [f_{ij} + f_{i+1,j} + f_{i,j+1}] \quad (d_0^- f)_{ij} = \frac{1}{3} [f_{ij} + f_{i-1,j} + f_{i,j-1}]$$

$$(d_1^+ f)_{ij} = [-f_{ij} + f_{i,j+1}] \quad (d_1^- f)_{ij} = [-f_{ij} + f_{i,j-1}]$$

$$(d_2^+ f)_{ij} = [2f_{ij} + f_{i+1,j} - 3f_{i,j+1}] \quad (d_2^- f)_{ij} = [2f_{ij} + f_{i-1,j} - 3f_{i,j-1}]$$

$$D_{00} = \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$$

$$(D_{00} f)_{ij} = \frac{1}{9} [f_{i-1,j+1} + f_{i,j+1} + f_{i+1,j} + 3f_{ij} + f_{i+1,j-1} + f_{i,j-1} + f_{i-1,j}]$$

$$D_{11} = 1, 1$$

$$d_1^+ = 1, \quad d_1^- = 1$$

$$(D_{11} f)_{ij} = -f_{i,j+1} + 2f_{ij} - f_{i,j-1}$$

$$D_{22} = 1, 1$$

$$d_2^+ = 1, \quad d_2^- = 1$$

$$(D_{22} f)_{ij} = -3f_{i-1,j+1} - 6f_{i,j+1} - 2f_{i+1,j} - 4f_{ij} + 2f_{i+1,j-1} - 6f_{i,j-1} - 3f_{i-1,j}$$

$$(d_0^+ f)_{ij} = \frac{1}{3} [f_{ij} + f_{i+1,j} + f_{i,j+1}]$$

$$(d_1^+ f)_{ij} = [-f_{ij} + f_{i,j+1}]$$

$$(d_2^+ f)_{ij} = [2f_{ij} + f_{i+1,j} - 3f_{i,j+1}]$$

9

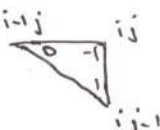
$$(d_0^- f)_{ij} = \frac{1}{3} [f_{ij} + f_{i-1,j} + f_{i,j-1}]$$

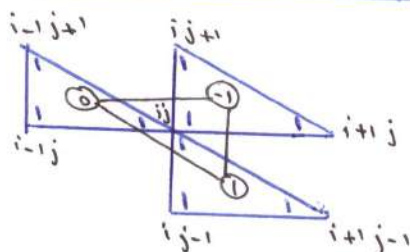
$$(d_1^- f)_{ij} = [-f_{ij} + f_{i,j-1}]$$

$$(d_2^- f)_{ij} = [2f_{ij} + f_{i-1,j} - 3f_{i,j-1}]$$

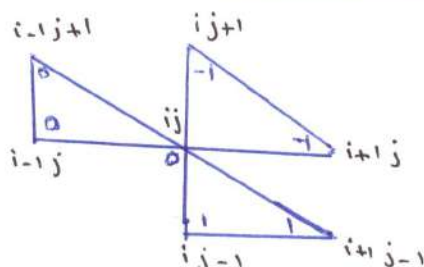
$$D_{01} f = \frac{1}{3} \cdot 1$$

$$d_0^+ = \frac{1}{3}$$


$$d_1^+ =$$


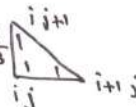


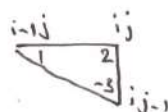
$$= \frac{1}{3}$$

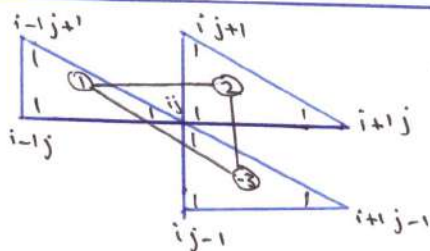


$$(D_{01} f)_{ij} = \frac{1}{3} [-f_{i,j+1} - f_{i+1,j} + f_{i,j-1} + f_{i+1,j-1}]$$

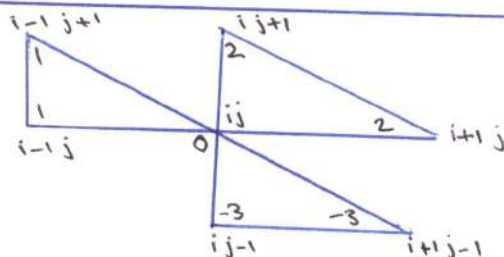
$$D_{02} f = \frac{1}{3} \cdot 1$$

$$d_0^+ = \frac{1}{3}$$


$$d_2^+ =$$




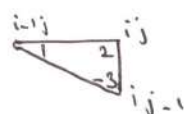
$$= \frac{1}{3}$$

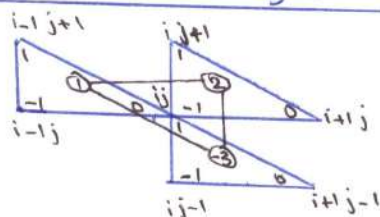


$$(D_{02} f)_{ij} = \frac{1}{3} [f_{i-1,j+1} + 2f_{i,j+1} + f_{i-1,j} + 2f_{i+1,j} - 3f_{i,j-1} - 3f_{i+1,j-1}]$$

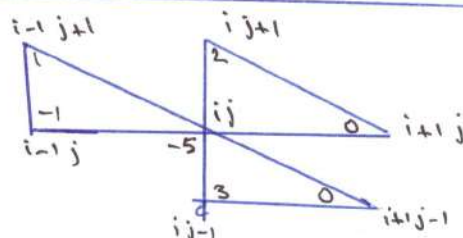
$$D_{12} f = 1 \cdot 1$$

$$d_1^+ = 1$$


$$d_2^- =$$


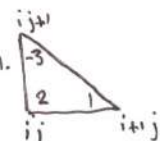


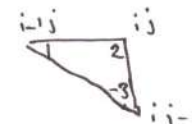
$$= 1$$

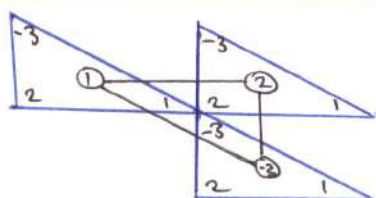


$$(D_{12} f)_{ij} = [f_{i-1,j+1} + 2f_{i,j+1} - f_{i-1,j} - 5f_{ij} + 3f_{i,j-1}]$$

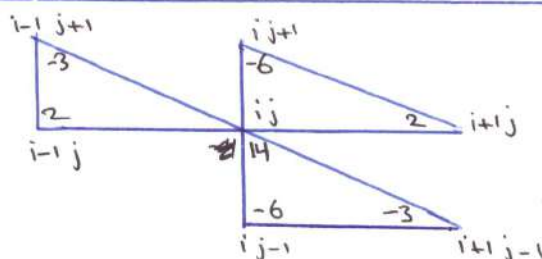
$$D_{21} f = 1 \cdot 1$$

$$d_2^+ = 1$$


$$d_2^- =$$




$$=$$



$$(D_{21} f)_{ij} = [-3f_{i-1,j+1} - 6f_{i,j+1} + 2f_{i-1,j} + 14f_{ij} + 2f_{i+1,j} - 6f_{i,j-1} - 3f_{i+1,j-1}]$$